

OMAR FARUK

Web Developer · Digital Marketer · AI / Computer Vision Researcher

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PROFESSIONAL SUMMARY

Computer Science & Engineering graduate (BUBT, 2026) with applied research in deep learning for medical imaging. Capstone thesis TumorMultiNet: a hybrid brain MRI classification and segmentation framework with Grad-CAM explainability, with a related paper accepted (IEEE) and a second paper submitted (ICCIT). Professional experience in web development, digital marketing and travel operations.

EDUCATION

B.Sc. in Computer Science & Engineering

Completed 2026

Bangladesh University of Business and Technology (BUBT), Dhaka

CGPA 3.24 / 4.00

Capstone thesis: TumorMultiNet (group project)

Coursework: Data structures and algorithms, object-oriented programming, database systems, software engineering, computer networks, web technologies.

Higher Secondary Certificate (Science)

2021

Kharrah Adarsha Degree Honours College

GPA 4.58 / 5.00

Secondary School Certificate (Science)

2019

Churain Tarini Bama High School

GPA 4.17 / 5.00

RESEARCH

TumorMultiNet: A Statistically Validated Hybrid Deep Learning Framework for Brain Tumor MRI Classification and Segmentation on BRISC2025 with Explainable AI

2026

Group capstone thesis · B.Sc. in Computer Science & Engineering · BUBT · 2026

- **Research areas:** Deep learning · Computer vision · Medical imaging · Brain MRI classification · Tumor segmentation · Explainable AI.
- **Final models:** classification EfficientNet-B0 + DenseNet121; segmentation SegFormer + DeepLabV3+; explainability Grad-CAM.
- **Data:** BRISC2025: 6,000 T1-weighted slices, 4 classes (glioma, meningioma, pituitary, no tumor), 4,793 pixel-level masks. External evaluation on the Cheng cohort: 3,064 slices from 233 patients.
- **Method:** Seven classification backbones and six segmentation architectures compared; the best two models of each branch combined into hybrids. Results checked with McNemar's exact test, the Wilcoxon signed-rank test and 95% confidence intervals.
- Delivered as a research prototype, not a clinical tool.

PUBLICATIONS

TumorMultiNet: A Statistically Validated Hybrid Deep Learning Framework for Brain Tumor MRI Classification and Segmentation on BRISC2025 with Explainable AI

Accepted · IEEE

Smart First Aid Box

Submitted · ICCIT

RESEARCH SKILLS

Core: Python · PyTorch · OpenCV · scikit-learn · Deep learning · Computer vision · Medical image analysis · Image classification · Segmentation · Explainable AI · Research writing

Libraries & methods: timm · torchvision · NumPy · Matplotlib · Statistical validation (McNemar, Wilcoxon, confidence intervals)

Environment: Kaggle (Tesla T4 GPU) · Overleaf · GitHub · VS Code · Microsoft Word & Excel

PROFESSIONAL EXPERIENCE

Web Developer & Digital Marketer

Current

Midtown Aabashon Ltd. · midtownaabashonltd.com

- Website development, digital marketing and brand-focused online execution.

Travel Operations & Ticketing Assistant

March 2026 – August 2026

Trip Fly BD · Dhaka, Bangladesh

- Air ticketing and reservation support on Sabre GDS; built website sections and the Smart Attendance System.

SELECTED PROJECTS

Smart Attendance System · Live · Internal tool for Trip Fly BD

Live app

Browser-based attendance system: rotating QR check-in, GPS verification, admin and employee panels, and a Google Sheets back end through Google Apps Script.

TRM Holidays · Live · Travel technology / business website

trmholidays.com

Travel agency website with tabbed flight, hotel and tour-package search, a domestic and international tour catalogue, a responsive layout and SEO foundations.

SKILLS & TOOLS

Web: HTML · CSS · JavaScript · TypeScript · Next.js · Responsive web development · Git & GitHub

Digital & business: Digital marketing · SEO foundations · Business automation · Workflow digitization · Sabre GDS

ACTIVITIES & LANGUAGES

BUBT ICPC 2025 (contest participant) · BIUPC Programming Contest (contest participant) · BUBT Innovtex Hackathon (event volunteer)

Languages: Bangla (native) · English (professional working proficiency)